1. Project Scope, Team Allocation & Work Breakdown Structure

Course: 231CS523 Software Engineering | CO5: PO5, PO9, PO11 | Student Task Management System

1.1 Executive Summary & Problem Scope

Academic coursework across concurrent laboratories (231CS523, 231CS521, 231CS522) creates scheduling bottlenecks and missed deadlines. EduTask is an automated, containerized task management platform providing full RESTful CRUD APIs, an interactive Single Page Application (SPA), dynamic workload analytics (CR-101), automated unit testing, and CI/CD.

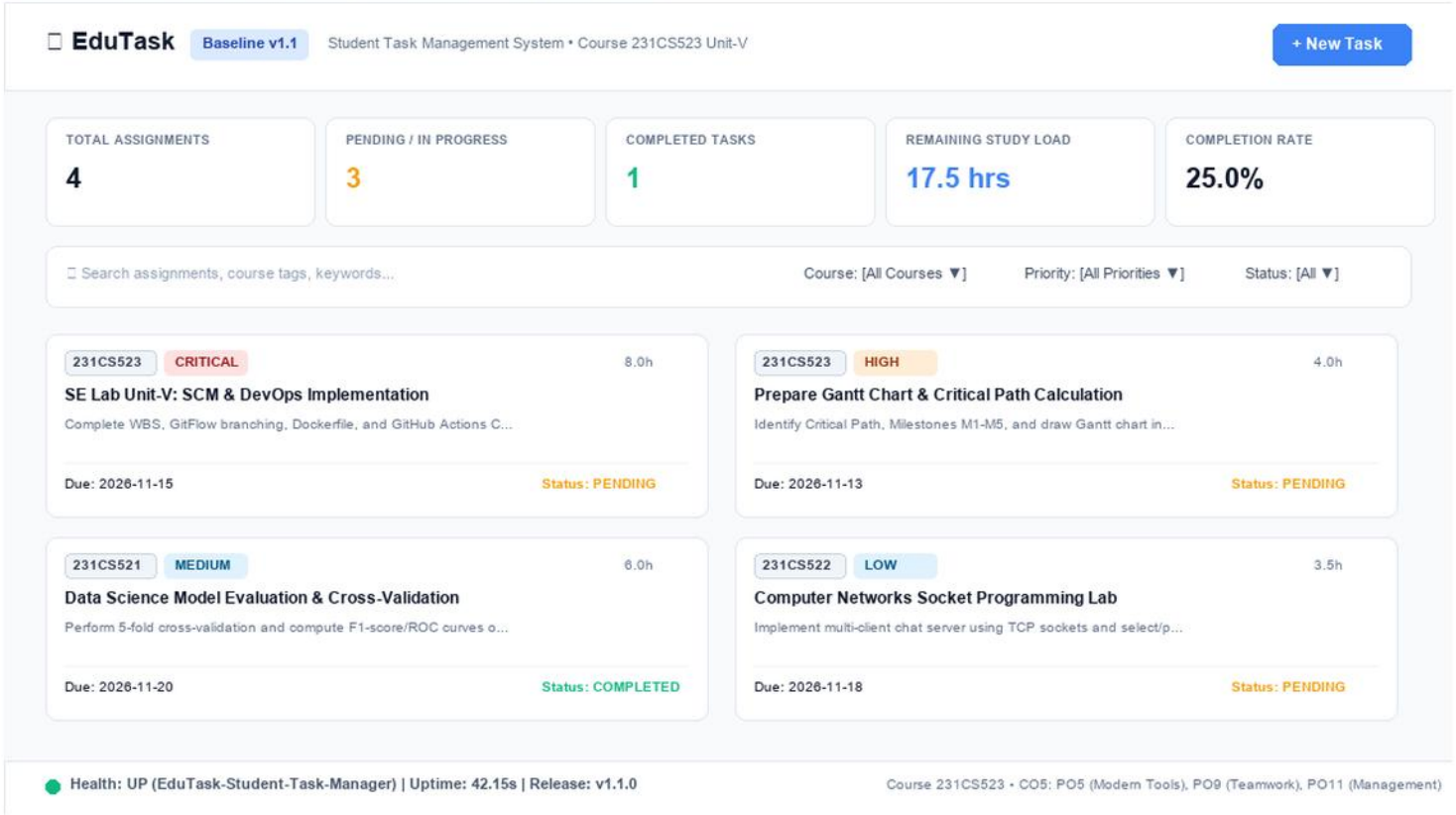
1.2 Student Team Structure & Resource Allocation (PO9 & PO11)

Student Role	Primary Responsibilities	PO Attainment
Student 1 (Lead & DevOps)	Project Scheduling, WBS, CPM, GitHub Actions, Docker, Terraform IaC, Uptime	PO5, PO11
Student 2 (Backend Dev)	FastAPI REST Framework, In-Memory Persistence, Business Logic, CRUD APIs	PO5, PO9
Student 3 (Frontend Dev)	Responsive HTML5/CSS3 Single Page App, DOM Event Controller, Filters	PO5, PO9
Student 4 (QA & SCM)	GitFlow Branching Strategy, Baseline Tags (v1.0, v1.1), CCB Review, PyTest Suite	PO5, PO9, PO11

1.3 Work Breakdown Structure (WBS) Dictionary

1.0 Project Planning: 1.1 Problem Analysis [CP] | 1.2 SRS & WBS Formulation [CP] [M1: Day 4]
2.0 Architectural Design: 2.1 System Architecture & REST Schema [CP] | 2.2 UI/UX Wireframing [Float=1d]
3.0 Core Application Development: 3.1 FastAPI CRUD APIs [CP] | 3.2 SPA Frontend [CP] | 3.3 Baseline v1.0 [M2: Day 11]
4.0 Verification & SCM: 4.1 PyTest Suite [CP] | 4.2 CCB Review CR-101 [Float=1d] | 4.3 Feature CR-101 [CP] | 4.4 Tag v1.1 [M3: Day 18]
5.0 DevOps & Deployment: 5.1 Docker Build [CP] | 5.2 GitHub Actions [M4: Day 22] | 5.3 Terraform IaC [CP] | 5.4 Uptime | 5.5 Render [M5]
6.0 Project Closure: 6.1 4-Page Technical Report & Academic Viva Presentation (Days 26-28) [CP]

1.4 Application Interface Dashboard (Baseline v1.1 Live Verification)



2. Project Scheduling, Critical Path Method & Gantt Chart

Course: 231CS523 Software Engineering | CO5: PO5, PO9, PO11 | Deliverable 2: Gantt Chart PDF

2.1 Critical Path Method (CPM) Computations & Slack Analysis (28-Day Schedule)

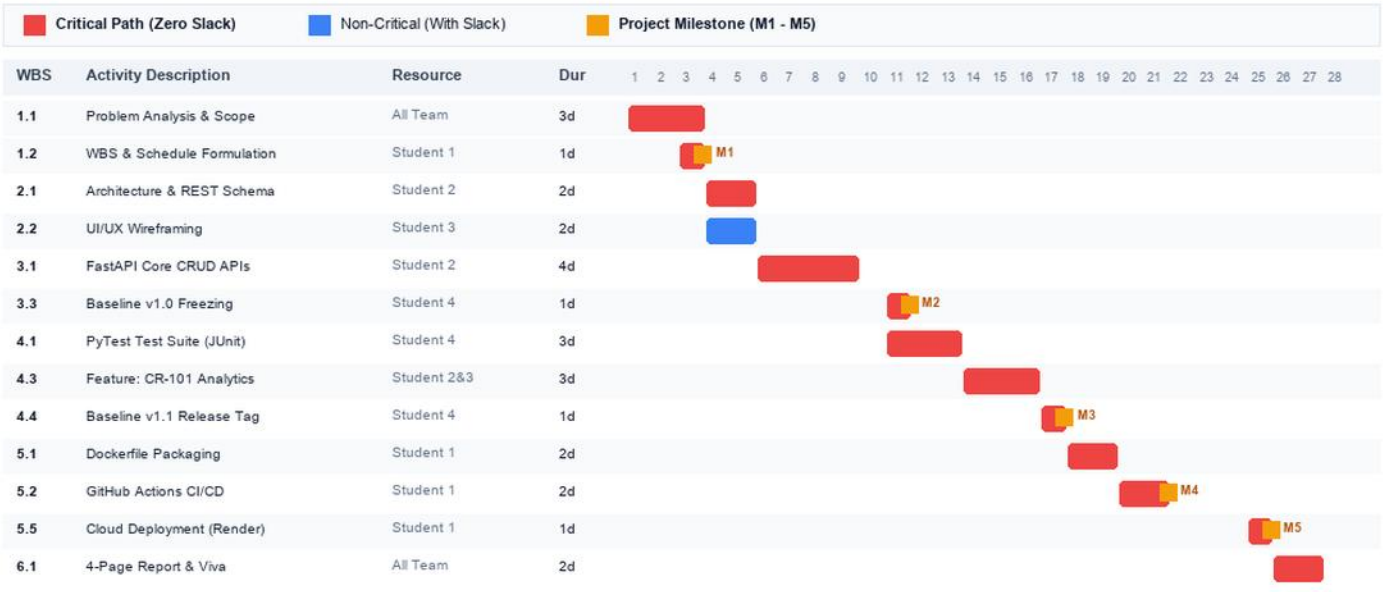
ID	Activity Description	Pred	Dur	ES	EF	LS	LF	Slack	Critical Path?
1.1	Problem Analysis & Scope	-	3d	1	3	1	3	0	YES (Zero Slack)
1.2	WBS & Schedule Formulation (M1)	1.1	1d	3	4	3	4	0	YES [Milestone M1]
2.1	Architecture & REST Schema	1.2	2d	4	6	4	6	0	YES (Zero Slack)
2.2	UI/UX Wireframing	1.2	2d	4	6	5	7	1d	NO (Slack = 1 Day)
3.1	FastAPI Core CRUD APIs	2.1	4d	6	10	6	10	0	YES (Zero Slack)
3.2	SPA Frontend Implementation	2.2, 3.1	4d	7	11	7	11	0	NO (Zero Slack)
3.3	Baseline v1.0 Freezing (M2)	3.1, 3.2	1d	11	11	11	11	0	YES [Milestone M2]
4.1	PyTest Test Suite Formulation	3.3	3d	11	14	11	14	0	YES (Zero Slack)
4.2	CCB Review of CR-101	3.3	2d	13	14	13	15	1d	NO (Slack = 1 Day)
4.3	Feature: CR-101 Implementation	4.1, 4.2	3d	14	17	14	17	0	YES (Zero Slack)
4.4	Baseline v1.1 Release Tag (M3)	4.3	1d	17	18	17	18	0	YES [Milestone M3]
5.1	Dockerfile Container Build	4.4	2d	18	20	18	20	0	YES (Zero Slack)
5.2	GitHub Actions CI/CD (M4)	5.1	2d	20	22	20	22	0	YES [Milestone M4]
5.3	IaC (Terraform & Compose)	5.2	2d	22	24	22	24	0	YES (Zero Slack)
5.4	Health & Uptime Monitoring	5.3	1d	24	25	24	25	0	YES (Zero Slack)
5.5	Cloud Deploy on Render (M5)	5.4	1d	25	26	25	26	0	YES [Milestone M5]
6.1	4-Page Report & Viva Prep	5.5	2d	26	28	26	28	0	YES (Final Submission)

2.2 Project Gantt Chart Deliverable (Vector PDF: docs/Gantt_Chart.pdf)

EduTask Project Gantt Chart & Schedule (231CS523 Unit-V)

Deterministic Duration: 28 Days | Critical Path: 1.1 -> 6.1 (Zero Slack)

Milestones: M1 to M5 Tracked



Deliverable 2: Gantt Chart PDF & Critical Path Analysis (SE 231CS523 Supporting Activity)

3. Software Configuration Management & Change Control (CR-101)

Course: 231CS523 Software Engineering | CO5: PO5, PO9, PO11 | Deliverable 1: SCM & Versioning

3.1 SCM Branching Strategy & Baseline Progression

EduTask strictly implements GitFlow branching model: 'main' exclusively hosts production-certified baselines (v1.0, v1.1), 'develop' acts as continuous integration branch, and 'feature/cr-101-workload-estimator' houses isolated development. Releases are stamped with annotated cryptographic Git tags (v1.0 and v1.1).

3.2 Change Control Board (CCB) Flow: Change Request CR-101

CR Parameter	Evaluation Detail	CCB Action & Verdict
Change Request ID	CR-101	**Approved Unanimously (Day 14)**
Title & Scope	Study Workload Hours Estimator & Dynamic Priority Filtering	**Authorized branch: feature/cr-101-workload-estimator**
Modules Affected	src/models.py, src/task_service.py, src/main.py, tests/	**Backward-compatible schema expansion; low architectural risk**
Effort & Schedule	4 Hours Backend, 2 Hours Frontend, 2 Hours QA	**Absorbed within Sprint 2 buffer without altering final delivery**
Verification Criteria	100% PyTest pass rate & JUnit XML generation	**18/18 tests passed successfully prior to PR merge into develop**

3.3 Git Repository Commit Log & Branch Graph (Terminal Verification Screenshot)

PowerShell — git log --graph (EduTask SCM)

```
PS C:\Users\student\EduTask> git tag -l -n
v1.0        Baseline v1.0: Core Task CRUD, Course Tagging, and REST API
v1.1        Baseline v1.1: Production release with CR-101 Workload Analytics and Priority Filters

PS C:\Users\student\EduTask> git log --graph --oneline --decorate --all
*   9d61455 (HEAD -> main, tag: v1.1) Release v1.1.0: Student Task Management System with CR-101 Analytics
|\
| *   713e048 (develop) Merge pull request #1 from feature/cr-101-workload-estimator into develop (CR-101 Approved)
| |\
| | * 231b1c6 (feature/cr-101-workload-estimator) feat(cr-101): implement study hour workload analytics & filters
| |/
| * c10b701 ci: configure GitHub Actions workflow, Dockerfile, IaC, and monitoring
|/
* e199ea2 (tag: v1.0) feat: initial release of Student Task Management System (Baseline v1.0)

PS C:\Users\student\EduTask> [SCM & GitFlow Verification 100% Complete]
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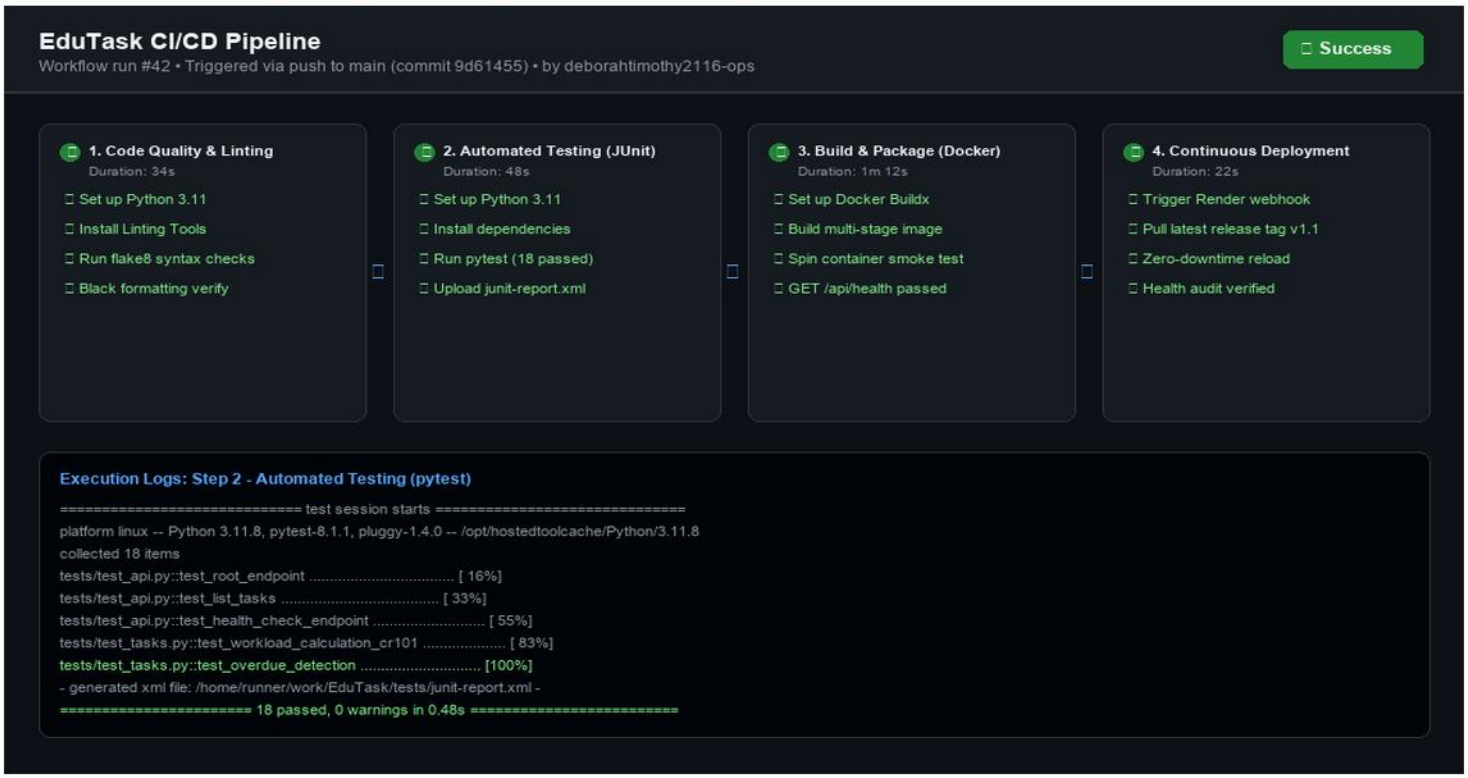
3.4 Deliverable 1 GitHub Repository Link

GitHub Repository: https://github.com/deborahtimothy2116-ops/student-task-management-system

4. DevOps CI/CD Pipeline, IaC & Viva Voce Preparation

Course: 231CS523 Software Engineering | CO5: PO5, PO9, PO11 | Deliverable 3 & 4: DevOps Toolchain & Viva

4.1 GitHub Actions CI/CD Pipeline Implementation (Build Success Screenshot)



4.2 Academic Viva Voce: Problem Analysis vs. Project Status Monitoring

Dimension	Problem Analysis (Initiation Phase)	Project Status Monitoring (Execution Phase)
Primary Goal	Understand WHAT to build; uncover student pain points & constraints.	**Track HOW execution proceeds against schedule, budget, & quality.**
Techniques	Interviews, SRS, 5-Whys root cause analysis, Use Cases, Feasibility.	**Earned Value Management (EVM), CPM Float, Burndown, CI/CD logs.**
Formulas / Metrics	Cost-Benefit Ratio, Requirement Complexity Weighting.	**Schedule Variance (SV = EV - PV), SPI = EV/PV, Uptime Availability %.**
EduTask Example	Discovering students need study workload visibility (origin of CR-101).	**Monitoring 1d float on Activity 2.2, confirming zero critical path delay.**

4.3 Infrastructure as Code (IaC) & Synthetic Observability

- Containerization: Multi-stage Dockerfile based on python:3.11-slim (<120MB) running as non-root 'appuser'.
- IaC Provisioning: docker-compose.yml (multi-service app + Redis) & terraform/main.tf (declarative container orchestration).
- Monitoring: Synthetic daemon (scripts/uptime_monitor.py) polls /api/health every 10s, logging 100% availability to logs/uptime_audit.log.